NARROGIN SENIOR HIGH SCHOOL
MATHEMATICS DEPARTMENT

MATH O MAGIC 3· 4 SOLUTIONS

1. YOU POP UP ALL NIGHT
2. OIL IT
3. YOUR STOMACH KEEPS TURNING
4. BECAUSE SHE WANTED TO TURN THEM INTO BRICKLAYERS
5. SO THE WATER WONT HIT PASSENGERS SQUARE IN THE EYE
6. A WHOLE NEW TWIST
7. HE LIKED TO CHOP AND CHANGE
8. BECAUSE HE THOUGHT IT WOULD BE JUST THE PLACE FOR A GOOD LOAF
9. A CHOP ASSISTANT
10. A POLICE DOG WORKING UNDER COVER
11. AN UDDER FAILURE
12. A TUBA TOOTHPASTE
13. A WET BLANKET
14. HOGWASH
15. JUST LOOK FOR THE GREY HARES
16. HE PICKED UP A HAMMER AND SAW
17. AN OLD TIMER
18. A RECEDING HARELINE
19. A HIPPOCHONDRIAC
20. LUMPY MILKSHAKES
21. SOME BRIGHT SPARK
22. A DOUGHNUT DISTURB SIGN
23. THEY DIDN'T KNOW HOW TO CONDUCT THEMSELVES
24. A TRIFLE DEAF
25. A CAMERA THAT TAKES PICTURES OF ITSELF
26. DURING WIZARDING HOURS
27. HE COULD FEEL IT IN HIS BONES
28. THEY FOUGHT TOOTH AND NAIL
29. THE FAIRIES TOOK ALL HER TEETH AWAY
30. YOU GET A BUZZY SIGNAL
31. BY ICICLE
32. TOOTH DEC AY
33. GROUND BEEF
34. POOCHED EGGS
35. A GET WELL CARD
36. JACK THE KIPPER
37. BECAUSE THEY CAN MAKE ENDS MEET
38. HE GOT STUCK IN THE MIDDLE OF A SUM
39. BALLET ACHES

MATH O MAGIC

3·4

by

Karen Anderson

399021 1

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What Do Dancers Get If They Overeat?

INSTRUCTIONS:
Graph each of the following pairs of equations on separate graph paper. Each pair will pass through a common point. Look for that point in the Code Key and notice the letter next to it. Write that letter next to the number of the question in the code below.

1. $y = x + 1$
$y = 2x$

2. $y = x + 4$
$y = 2x + 5$

3. $y = x + 2$
$y = 2x$

4. $y = x + 7$
$y = 3x + 11$

5. $y = x + 4$
$y = 4 - x$

6. $y = x - 3$
$y = 2x - 3$

7. $y = x + 5$
$y = 2x + 8$

8. $y = x - 2$
$y = 3x + 2$

CODE KEY	
(– 3, 2) –	S
(– 1, – 1) –	R
(1, 2) –	L
(– 2, – 4) –	B
(3, 2) –	M
(– 2, 5) –	A
(0, 1) –	O
(0, 4) –	C
(– 2, 2) –	W
(– 1, 3) –	E
(1, 4) –	N
(2, 3) –	I
(0, – 3) –	H
(2, 4) –	T

8	
4	
1	
1	
2	
3	
4	
5	
6	
2	
7	

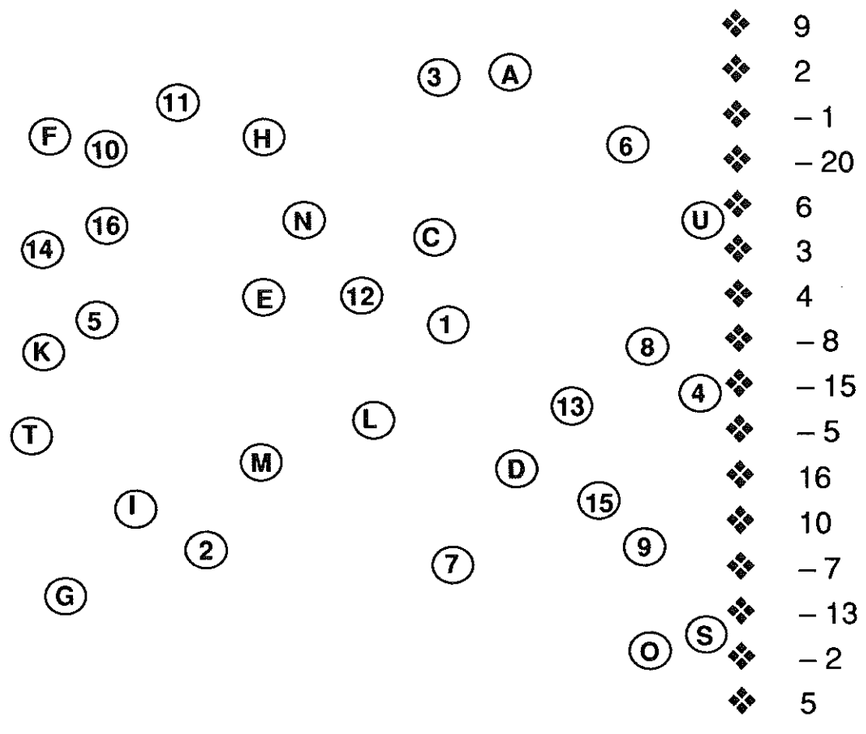


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WHAT HAPPENED TO THE BOY WHO DID HIS HOMEWORK WITH STICKY HANDS?

Solve each of the following equations. Look for the answers in the right hand column. Rule a line to connect the equation with its solution. Each line will pass through a number and a letter. Write the letter above its matching number in the code.

2x + x + 6 = 18	9
4x + 2x + 9 = 21	2
6x + 3x − 5 = 22	−1
9x + 2x − 15 = 95	−20
3x + 6 + 4x = 41	6
4x + x + 6 = −4	3
3x + 5 + x = 1	4
7x + 6 + 2x = −39	−8
2x − 7 + x = −28	−15
7x − 3x − 5 = −37	−5
4x − 2x = −26	16
5x − 4x − 3 = 3	10
2x − 5x = −27	−7
5x − 7x = −32	−13
2x − 4x + 10 = 40	−2
3x − 6x + 12 = 72	5



11	16	15	14	8	5	8	1	12	13	7	6

8	11	16	9	7	4	4	2	16	14	10	3	5	1	9

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INTRODUCTION

MATH O MAGIC 3·4 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.
2. INCENTIVE – As a reward for students who finish their work early.
3. SELF CHECKING – For students to be able to check their own work as they go along.
4. EXTRA PRACTICE – A fun way of doing extra practice on skills development.
5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.
6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

WHAT HAPPENS IF YOU PLUG YOUR ELECTRIC BLANKET INTO THE TOASTER?

Decide whether the following numbers are prime or composite. Shade in the circle above each **PRIME** number. You will be left with 16 letters. Write those letters in order from left to right and you will have the answer to the riddle.

M	Y	A	O	U	N	L	S	P	R
3	9	11	12	16	17	23	29	30	31

A	O	S	P	U	R	P	A	L	E
37	38	41	45	48	53	57	60	65	67

L	N	I	A	G	H	T	S
68	72	87	89	91	93	95	97

_____ _____ ____ _____ _____

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Why are contortionists good at budgeting?

Solve each of these equations. Look for your answers in the grid below. Cross out each box containing a correct solution. When you finish there will be 15 boxes remaining. Write the letters in those boxes in order from left to right.

$2x + 6 = 16$
$8x + 16 = 64$
$2x - 7 = -23$
$90 - 4x = 6$
$94 - 2x = 54$

$3x + 5 = 17$
$5x + 14 = -1$
$7x + 5 = -100$
$40 - 3x = 67$
$20 - 8x = 52$

$5x + 7 = 22$
$7x + 6 = 20$
$10x - 15 = -85$
$100 - 9x = 199$
$120 - 6x = 24$

AN x = 5	BE x = 10	CA x = 1	RS x = – 8	ST x = 3
US x = –5	AI x = -11	ET x = –20	HE x = –10	RS x = – 4
YC x = 25	ON x = 16	RM x = 21	AN x = 0	GH x = 2
MA x = – 13	EN x = 6	KE x = -22	CK x = - 7	BO x = 18
TH x = 9	EN x = 12	OA x = 20	LK x = 4	DS x = – 1
EA x = – 3	OR x = – 15	ME x = – 6	ST x = – 9	ET x = – 2

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WHAT FISH ARE OTHER FISH MOST FRIGHTENED OF?

INSTRUCTIONS
For each of the following linear functions, determine the vertical intercept. Find that point in the code below. Write the letter of each question above its vertical intercept.

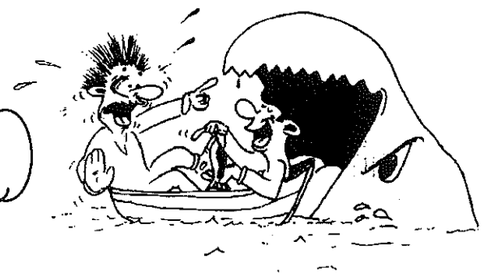
E $y = x + 2$

C $y = x - 3$

I $y = x + 1$

T $y = 2x - 1$

H $y = 6x - 6$



J $y = 3x - 2$

P $y = x + 4$

R $y = x + 3$

A $y = 4x - 5$

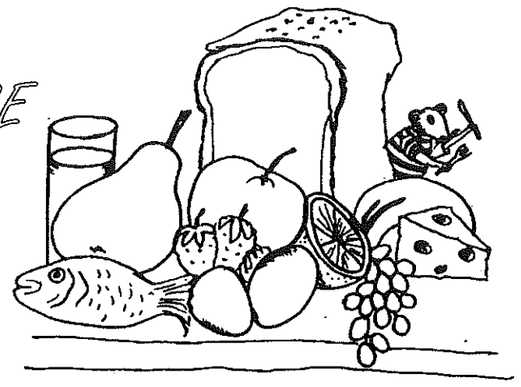
K $y = 2x - \frac{1}{2}$

$(0, -2)$	$(0, -5)$	$(0, -3)$	$(0, -\frac{1}{2})$	$(0, -1)$	$(0, 6)$	$(0, 2)$

$(0, -\frac{1}{2})$	$(0, 1)$	$(0, 4)$	$(0, 4)$	$(0, 2)$	$(0, 3)$

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WHAT SHOULD YOU DO WHEN YOU HEAR A MOUSE SQUEAKING IN THE MIDDLE OF THE NIGHT?



Simplify each of the following. Look for your answers in the grid below. Shade in each square containing a correct answer. When you finish you will have the answer to the riddle.

$8 - (-12)$
$-3 - (-12)$
$-6 - 7 - 2$
$7 - 9 - (-7)$
$-24 - 13 - 10$

$6 - 10$
$5 - (-5)$
$-5 - 8$
$5 - 7 + (-10)$
$20 - (-8) - (-8)$
$44 + (-13) + (-5)$

$-3 + 9$
$12 - 41$
$9 - (-15)$
$9 - 16 - 14$

$15 - 31$
$16 - 17$
$15 - (-10) + 8$
$-8 - 3 - (-3)$
$22 - (-6) - 5$
$17 - (-18) + (-6)$

$10 + (-13) + (-15)$
$-8 + 6$
$-3 + 5 - 8$







−29	29	5	0	−6	−25	−1	−11	16
−15	−35	−13	7	26	−28	23	−20	−7
−47	6	−12	3	33	46	−4	36	20
11	17	−22	21	−36	19	14	−3	2
42	−41	9	40	−8	−18	−16	−31	41
4	−5	10	1	−17	24	−14	1	13
18	−19	−21	8	35	−2	−10	15	−9



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WHAT HAPPENS WHEN YOU SWALLOW A DOORKNOB?

36	25	49	45	-80	-49	25	-28	-18	-24	-42	

30	-30	-30	-12	-80	-49	49	45	-72	24	-72	-16

Simplify the following. Look for your answers in the code above. Write the letter of each question above its answer.

C	(– 8) x 3	**I**	(- 6) x (- 4)
T	7 x (– 7)	**O**	(- 5) x (- 5)
A	3 x (– 6)	**N**	(- 9) x 8
R	(- 5) x (– 9)	**G**	8 x (-2)
M	4 x (– 7)	**U**	(-7) x (- 7)
E	5 x (– 6)	**Y**	(- 4) x (- 9)
S	8 x (-10)	**P**	2 x (- 6)
H	(- 6) x 7	**K**	(- 5) x (- 6)

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WHAT DO YOU SEND A PERSON WHO IS SEARCHING FOR OIL?

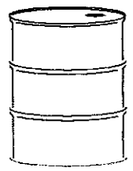


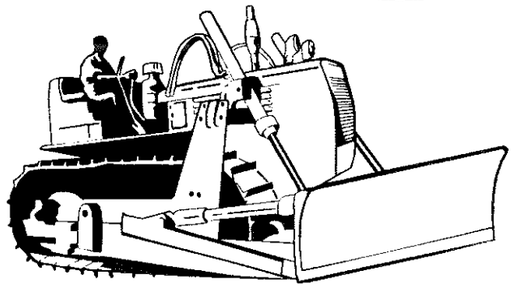
INSTRUCTIONS:
*Determine the **gradient** of each of these linear functions. Look for your answer in the Code Key and notice the letter next to it. Write the letter above the number of the question in the puzzle below.*

1. $y = 2x + 3$
2. $y = x + 1$
3. $y = -x + 4$
4. $y = -2x + 2$
5. $y = 3x + 1$
6. $y = 4 - 3x$
7. $y = \frac{1}{2}x + 4$
8. $y = \frac{x}{3} + 2$
9. $y = 6 - 4x$

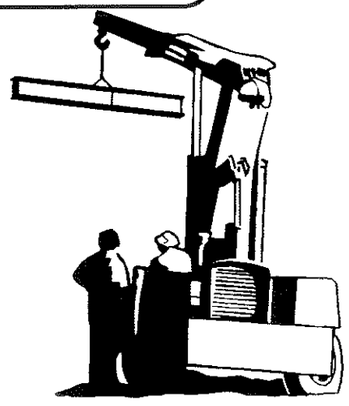
CODE KEY

-3	:	T
1	:	A
4	:	O
-1	:	W
$\frac{1}{2}$	:	E
-4	:	R
$-\frac{1}{2}$	:	N
2	:	G
$-\frac{1}{3}$	:	M
-2	:	D
$\frac{1}{3}$	:	C
3	:	L





2	1	7	6	3	7	5	5	8	2	9	4



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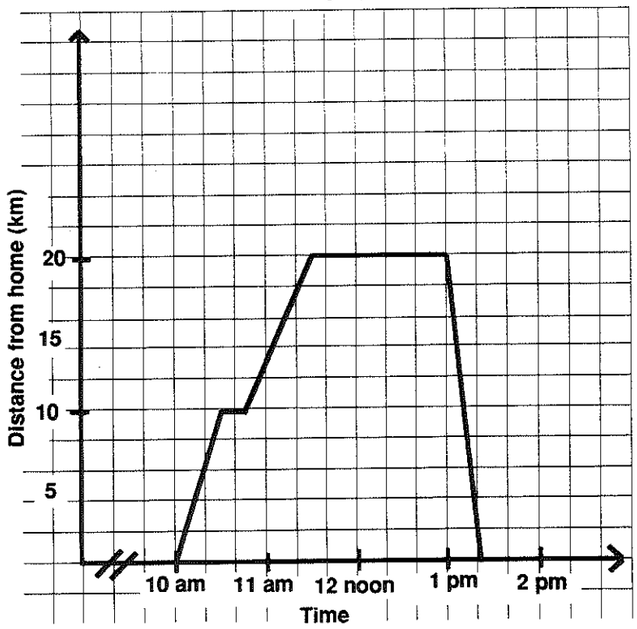


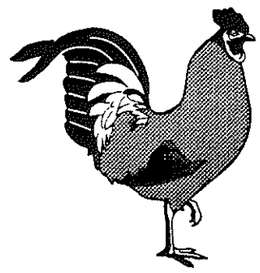
WHAT DO YOU GET WHEN YOU CROSS A DOG WITH A HEN?

INSTRUCTIONS:
Examine the travel graph below and answer the questions that follow. Look for your answers in the Code Key and notice the letters next to them. Write the letter above the number of the question in the puzzle at the top of the page.

5	7	7	3	4	6	8	6	1	1	2

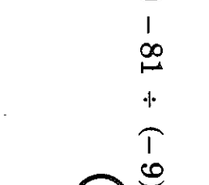
Lyn went cycling last weekend. After riding for half an hour she discovered a flat tyre. She mended the puncture and then cycled off to a friend's house. After having lunch with her friend, she was given a lift home by car. Shown below is a travel graph of Lyn's journey.





1. How far did Lyn ride before she got the flat tyre?
2. What was Lyn's cycling speed on this first leg of the trip?
3. How long did it take to mend the puncture?
4. What was her cycling speed on the second leg from the break to her friend's house?
5. How many hours did she spend at her friend's house?
6. At what speed did the car travel?
7. What total distance did Lyn cover by bike and car?
8. What was Lyn's total travelling time, in minutes, (excluding breaks)?

N	30 km/hr	A	15 minutes
P	1·5 hours	K	13 km/hr
C	20 minutes	O	40 km
G	10 km	E	60 km/hr
T	15 km	L	3 hours 20 minutes
M	40 km/hr	H	15 km/hr
S	20 km/hr	D	90 minutes



WHY DID THE WOMAN FEED HER CHICKENS CEMENT?

INSTRUCTIONS:
Simplify each of the following. Look for your answers in the code . Write the letter of the question above the answer. Keep working until you have the answer to the riddle.

| | | | | | | | | | |

WHY DO SHIPS HAVE ROUND PORTHOLES?

Simplify each of the following. Find your answers in the code below. Above each answer, write the letter of that question.

H $(-6) \times (-4)$

G $(-21) \div (3)$

O $45 \div (-9)$

R $(-2) + (-5) + (-6)$

Q $(-8) - (-12)$

T $20 + (-13)$

W $(-9) \times (-9)$

A $(-33) \div (-11)$

I $(-8) - 10 + (-12)$

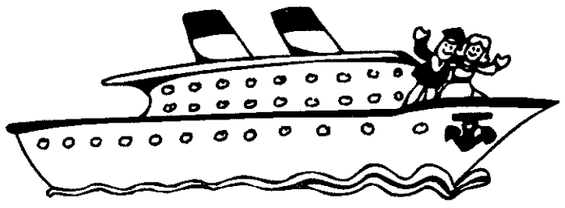
N $9 \times (-6)$

S $15 - (-8) - (-2)$

U $(-36) \div 6$

P $9 - (-11) - 4$

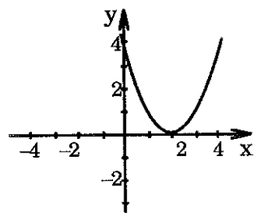
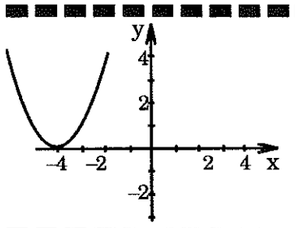
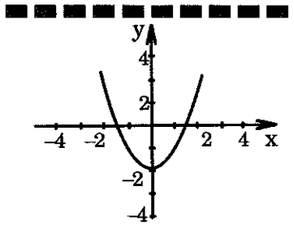
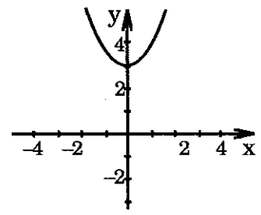
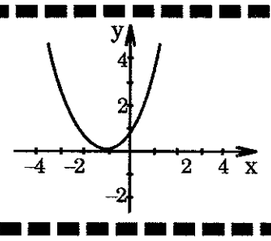
E $20 + (-3) + (-5)$



Y $(2) \times (-7)$

25	*–5*	*7*	*24*	*12*	*81*	*3*	*7*	*12*	*–13*	*81*	*–5*	*–54*	*7*	
24	*–30*	*7*	*16*	*3*	*25*	*25*	*12*	*–54*	*–7*	*12*	*–13*	*25*		
25	*4*	*–6*	*3*	*–13*	*12*	*–30*	*–54*	*7*	*24*	*12*	*12*	*–14*	*12*	

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6		$y = (x+2)^2$ **M**	$y = x^2 - 2$ **D**	$y = (x-2)^2$ **G**
7		$y = x^2 + 4$ **A**	$y = (x-4)^2$ **I**	$y = (x+4)^2$ **E**
8		$y = x^2 - 2$ **U**	$y = (x-2)^2$ **O**	$y = x^2 + 2$ **E**
9		$y = (x-3)^2$ **T**	$y = (x+3)^2$ **S**	$y = x^2 + 3$ **R**
10		$y = (x-1)^2$ **T**	$y = (x+1)^2$ **D**	$y = (x-1)^2$ **C**

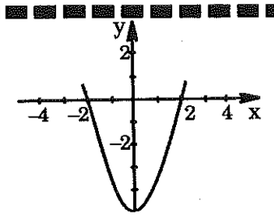
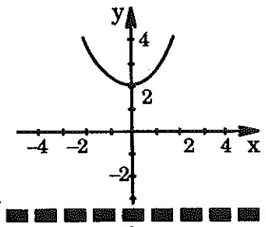
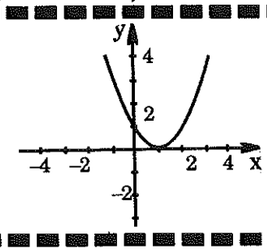
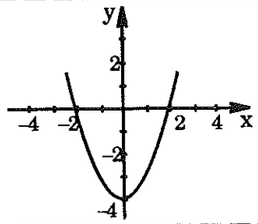
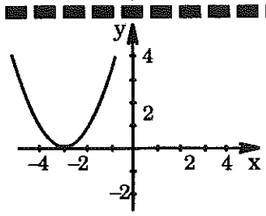
WHAT DO YOU CALL CATTLE THAT SIT ON THE GRASS?

6 9 4 8 3 10 5 1 7 2

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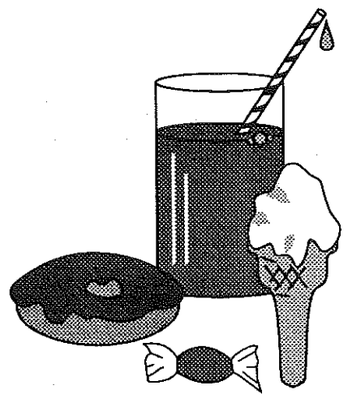
WHAT DO YOU CALL CATTLE THAT SIT ON THE GRASS?

INSTRUCTIONS:
Shown below are some graphs. Circle the equation for each function from the three choices given in the right hand columns. Notice the letter next to the circled equation. Write the letter above the number of the question in the code below.

1 	$y = x^2 - 5$	$y = x^2 + 5$	$y = (x-5)^2$
	E	A	U
2 	$y = x^2 - 2$	$y = x^2 + 2$	$y = (x+2)^2$
	M	F	G
3 	$y = (x-1)^2$	$y = x^2 + 1$	$y = x^2 - 1$
	N	S	T
4 	$y = x^2 + 4$	$y = (x-4)^2$	$y = x^2 - 4$
	R	A	O
5 	$y = x^2 + 9$	$y = x^2 + 3$	$y = (x+3)^2$
	R	G	B

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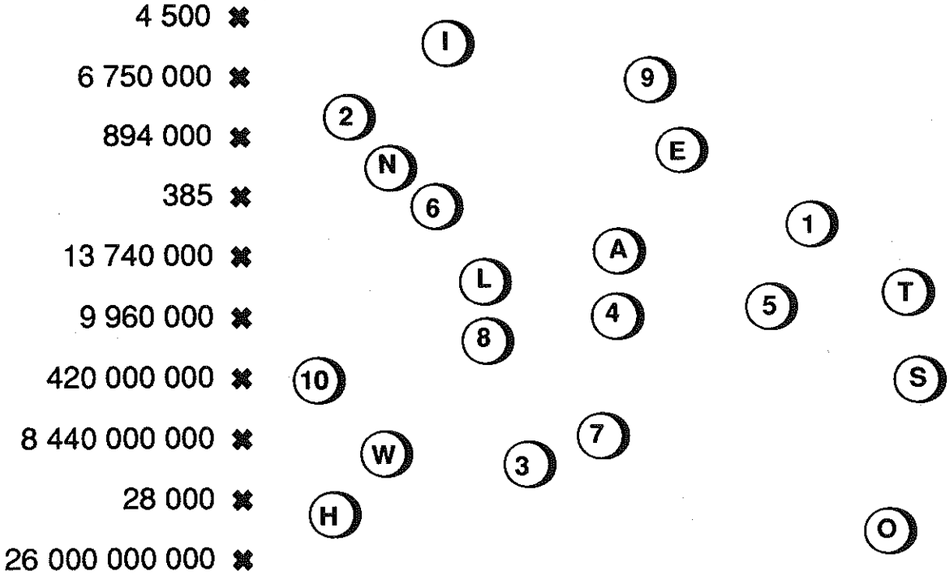
continued on next page



WHAT WOULD YOU GET IF YOU CROSSED A DOUGHNUT WITH A PRETZEL?

INSTRUCTIONS:
Match each number in decimal form with its scientific form. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the appropriate number in the puzzle below?

4 500 ✖	✖ $2{\cdot}8 \times 10^4$
6 750 000 ✖	✖ $1{\cdot}374 \times 10^7$
894 000 ✖	✖ $4{\cdot}5 \times 10^3$
385 ✖	✖ $2{\cdot}6 \times 10^{10}$
13 740 000 ✖	✖ $8{\cdot}94 \times 10^5$
9 960 000 ✖	✖ $6{\cdot}75 \times 10^6$
420 000 000 ✖	✖ $4{\cdot}2 \times 10^8$
8 440 000 000 ✖	✖ $3{\cdot}85 \times 10^2$
28 000 ✖	✖ $8{\cdot}44 \times 10^9$
26 000 000 000 ✖	✖ $9{\cdot}96 \times 10^6$



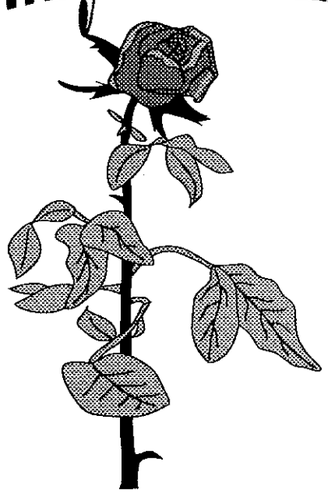
8	3	5	7	4	6	1	6	3

2	3	9	10	2



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Why did Henry the VIII have so many wives?



5	13	7	9	1	13	4	2	12				

8	5	12	11	3	6	4	8	5	3	6	10	13

Match each number written in scientific notation with its decimal equivalent. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the appropriate number in the code above.

5×10^{-7}	0·7
$7{\cdot}45 \times 10^{-3}$	0·000 95
$4{\cdot}26 \times 10^{-2}$	0·083
$3{\cdot}7 \times 10^{-3}$	0·000 885
$8{\cdot}3 \times 10^{-2}$	0·003 7
$8{\cdot}85 \times 10^{-4}$	0·007 45
$4{\cdot}8 \times 10^{-5}$	0·06
7×10^{-1}	0·000 009 8
$9{\cdot}25 \times 10^{-1}$	0·000 000 5
$9{\cdot}8 \times 10^{-6}$	0·925
6×10^{-2}	0·000 048
$9{\cdot}5 \times 10^{-4}$	0·001 5
$1{\cdot}5 \times 10^{-3}$	0·042 6

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Graph these quadratic equations on the grid below. Each graph will pass through two of the four given points. Cross out the two boxes containing correct coordinates. When you finish, ten boxes will remain. Rearrange the letters in each box in numerical order to give the answer to the riddle.

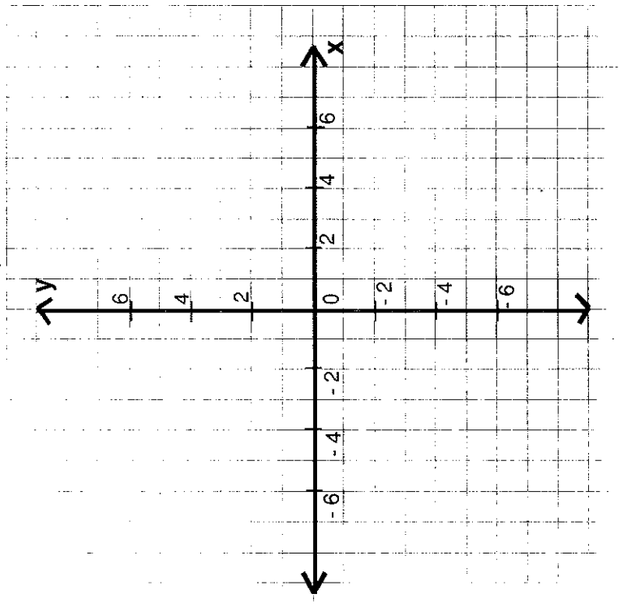
$y = x^2 - 4$ $y = (x-2)^2$ $y = (x+5)^2$

$y = x^2 + 3$ $y = -(x-4)^2$

$y = x^2 - 4$	4 – T (0, 4)	6 – R (0, –4)	3 – 0 (2, 0)	9 – A (4, 0)
$y = x^2 + 3$	2 – E (0, 3)	8 – L (1, 4)	10 – Y (1, 3)	2 – 0 (0, –3)
$y = (x-2)^2$	1 – S (0, 4)	5 – H (–2, 0)	9 – T (2, 0)	7 – E (2,4)
$y = (x+5)^2$	10 – N (–3, 4)	6 – D (5, 0)	1 – T (0, 5)	4 – M (–5, 0)
$y = -(x-4)^2$	3 – O (–4, 0)	8 – C (0, 1)	7 – A (4, 0)	5 – G (3, –1)

1	2	3	4	5	6	7	8	9	10

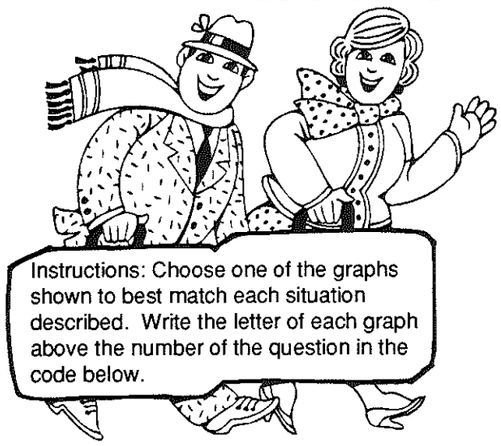
What Ailment Do Vampires Fear the Most?



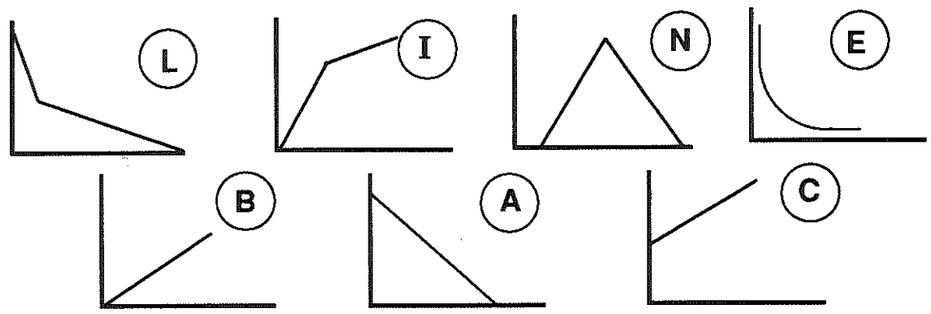
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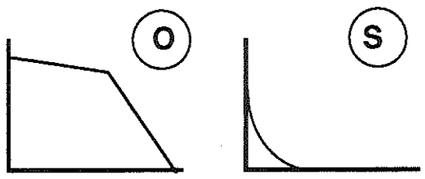
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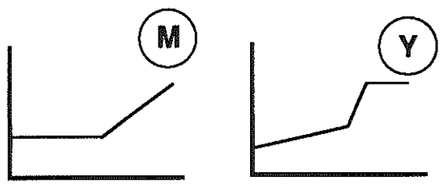
HOW DOES JACK FROST TRAVEL?



Instructions: Choose one of the graphs shown to best match each situation described. Write the letter of each graph above the number of the question in the code below.



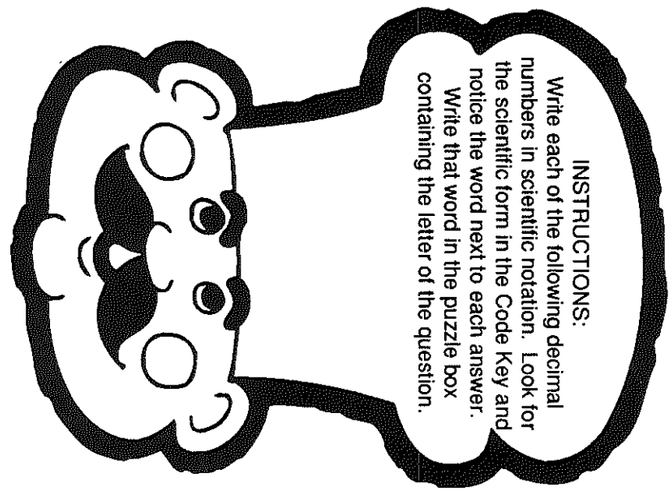




1. A taxi charges a flagfall of $5, plus an extra 50 c per 200 metres. Consider the graph for "**cost of taxi - distance travelled.**"
2. A lotto prize is won by a group. Consider the graph for "**individual prize - number of people in group.**"
3. A tree planted last year grew very slowly for the first 6 months, very rapidly during the next 3 months and did not increase its height at all during the following 3 months. Consider the graph for "**height - time.**"
4. The more pens Carol buys, the greater the total cost will be. Consider the graph for "**number - cost.**"
5. Shane was driven to his friend's house and then he and his friend walked the remaining distance to school. Consider the graph for "**distance from home - time.**"
6. A parachutist free-fell for 15 seconds after jumping from a plane. They then opened the parachute and fell slowly to the ground. Consider the graph for "**height - time.**"

4	**3**	**5**	**1**	**5**	**1**	**6**	**2**

WHY DID THE LAZY MAN TRY TO GET A JOB WITH A BAKER?



INSTRUCTIONS:
Write each of the following decimal numbers in scientific notation. Look for the scientific form in the Code Key and notice the word next to each answer.
Write that word in the puzzle box containing the letter of the question.

A		**B**		**C**			
D		**E**		**F**			
G		**H**		**I**			
J		**K**		**L**			
M							

CODE KEY

5×10^{-5}	A
$8{\cdot}205 \times 10^{7}$	WOULD
$5{\cdot}78 \times 10^{-2}$	START
$4{\cdot}4 \times 10^{-2}$	LOAF
$7{\cdot}14 \times 10^{5}$	BECAUSE
$4{\cdot}64 \times 10^{5}$	BREAD
$6{\cdot}65 \times 10^{5}$	FOR
$1{\cdot}408 \times 10^{8}$	THE
$6{\cdot}58 \times 10^{6}$	HE
$2{\cdot}75 \times 10^{-4}$	DOUGH
$9{\cdot}5 \times 10^{-1}$	JUST
4×10^{-2}	THOUGHT
$1{\cdot}59 \times 10^{4}$	GOOD
$3{\cdot}67 \times 10^{-2}$	SOGGY
$7{\cdot}2 \times 10^{-3}$	PLACE
$3{\cdot}6 \times 10^{-4}$	BE
$3{\cdot}75 \times 10^{-1}$	IT

A	714 000
B	6 580 000
C	0·04
D	0·375
E	82 050 000
F	0·000 36
G	0·95
H	140 800 000
I	0·007 2
J	665 000
K	0·000 05
L	15 900
M	0·044

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WHAT IS THE DEFINITION OF A BUTCHER'S APPRENTICE?

INSTRUCTIONS:
Simplify the following. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of each question in the puzzle below.

1. $\frac{1}{2} + \frac{1}{3}$
2. $\frac{1}{4} + \frac{1}{2}$
3. $\frac{2}{5} + \frac{1}{2}$
4. $\frac{5}{8} - \frac{1}{4}$
5. $\frac{4}{5} - \frac{1}{2}$
6. $\frac{5}{6} - \frac{1}{2}$
7. $\frac{9}{10} - \frac{1}{2}$
8. $\frac{1}{3} + \frac{1}{4}$
9. $\frac{2}{5} + \frac{1}{3}$

CODE KEY

A	$\frac{7}{12}$
O	$\frac{3}{8}$
P	$\frac{5}{6}$
N	$\frac{2}{5}$
H	$\frac{11}{15}$
T	$\frac{3}{4}$
I	$\frac{1}{3}$
S	$\frac{3}{10}$
C	$\frac{9}{10}$

8	3	9	4	1	8	5	5	6	5	2	8	7	2	

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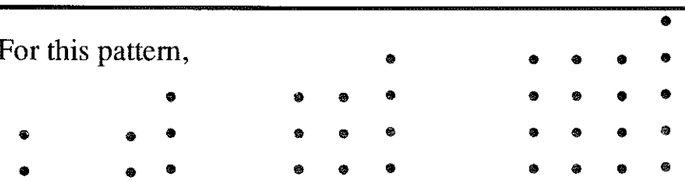
WHAT HAPPENS WHEN YOU TELEPHONE A BEE?

28	26	65	226	39	199	7

12	65	13	13	28	4	22	226	8	7	1 228

Answer the following questions. Look for your answers in the code above. Write the letter of each question above its answer.

For this pattern,



the rule for finding the number of dots, d, given the number of dots on the base, b, is $d = b^2 + 1$

How many dots are there in patterns with
- O 5 dots on the base?
- U 8 dots on the base?
- G 15 dots on the base?

How many dots are on the base if there is a total of
- S 17 dots?
- A 50 dots?
- Z 170 dots?
- I 485 dots?

For this pattern, the rule for finding the number of dots, d, given the number of dots on the base, b, is $d = b^2 + 3$



How many dots are there in patterns with
- E 6 dots on the base?
- T 14 dots on the base?
- L 35 dots on the base?

How many dots are on the base if there is a total of
- N 67 dots?
- B 147 dots?
- Y 787 dots?



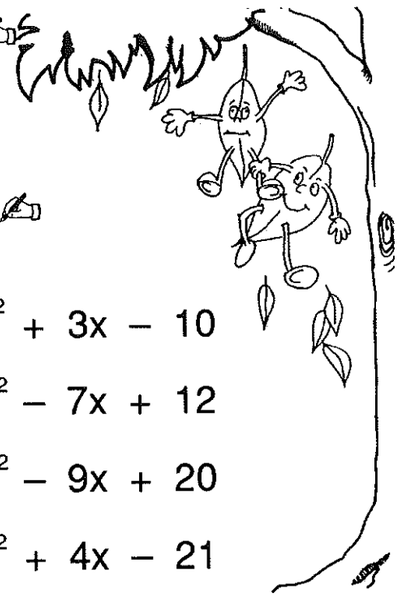
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WHAT HAPPENED WHEN THE GIRL WENT TO SLEEP WITH HER HEAD UNDER THE PILLOW?

8	7	10	3	6	13	12	13	10	2	8	9	9	5

6	1	1	7	10	12	8	10	10	8	7	6	11	6	4

Factorize each of these trinomials. Look for the factors in the grid below. Cross out each box containing a correct pair. When you finish there will be 13 boxes remaining. Write the letter in each box above the appropriate number in the code box.



$x^2 + 6x + 8$

$x^2 - 3x + 2$

$x^2 + 7x + 12$

$x^2 + 9x + 18$

$x^2 + 3x - 10$

$x^2 - 7x + 12$

$x^2 - 9x + 20$

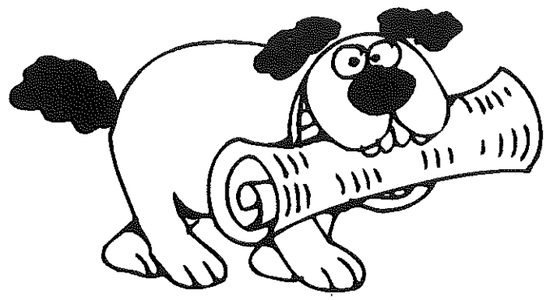
$x^2 + 4x - 21$

$x^2 - 8x + 12$

$x^2 + 7x + 10$

3 – F (x + 1)(x + 6)	1 – M (x – 8)(x + 4)	7 – H (x + 5)(x + 4)	5 – M (x + 5)(x + 2)	2 – T (x – 6)(x – 2)
9 – A (x – 3)(x – 4)	13 – I (x – 4)(x – 6)	6 – A (x – 2)(x + 4)	4 – W (x – 5)(x – 4)	1 – L (x – 2)(x – 3)
3 – G (x – 2)(x – 1)	10 – E (x + 5)(x – 3)	12 – S (x + 6)(x – 5)	5 – K (x + 9)(x + 2)	9 – O (x + 3)(x + 3)
4 – Y (x + 4)(x – 6)	11 – N (x + 6)(x + 3)	11 – W (x – 3)(x – 5)	8 – T (x – 2)(x – 2)	7 – N (x – 2)(x + 5)
8 – R (x + 2)(x + 4)	12 – R (x + 7)(x + 2)	10 – U (x + 7)(x – 3)	2 – S (x + 1)(x – 2)	6 – U (x + 4)(x + 3)

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What sort of dog says: "Meow"?

15	3	6	14	7	12	5	1	6	9

11	6	13	2	7	8	9	10	8	1	5	13

12	6	4	5	13

INSTRUCTIONS:
Simplify each of the following. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle.

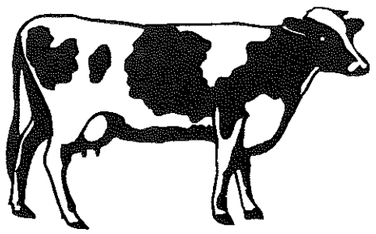
1. $1\frac{1}{2} + 1\frac{1}{3}$
2. $2\frac{1}{2} - 1\frac{1}{4}$
3. $3\frac{2}{3} - 1\frac{1}{2}$
4. $1\frac{3}{4} + \frac{1}{2}$
5. $\frac{4}{5} + \frac{1}{2}$
6. $\frac{2}{3} + \frac{3}{4}$
7. $2\frac{3}{4} - 1\frac{2}{3}$
8. $\frac{9}{10} + \frac{1}{2}$
9. $3\frac{7}{10} - 1\frac{2}{5}$
10. $1\frac{3}{4} + 1\frac{3}{8}$
11. $4\frac{3}{4} - 1\frac{1}{2}$
12. $2\frac{1}{2} - 1\frac{2}{5}$
13. $1\frac{7}{8} + \frac{3}{4}$
14. $\frac{3}{5} + 1\frac{1}{2}$
15. $2\frac{1}{2} - 1\frac{1}{3}$

CODE KEY

C	$1\frac{1}{10}$
A	$1\frac{1}{6}$
K	$1\frac{1}{4}$
D	$2\frac{5}{6}$
W	$3\frac{1}{4}$
N	$1\frac{2}{5}$
E	$1\frac{3}{10}$
I	$1\frac{1}{12}$
V	$2\frac{1}{4}$
G	$2\frac{3}{10}$
P	$2\frac{1}{6}$
L	$2\frac{1}{10}$
O	$1\frac{5}{12}$
R	$2\frac{5}{8}$
U	$3\frac{1}{8}$

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WHAT DO YOU CALL A COW THAT WON'T GIVE MILK?



3	5	2	4	4	9	7

8	3	6	1	2	7	9

INSTRUCTIONS:
Simplify the following problems. Look for your answers in the right hand column. Rule a line to connect the question with its answer. Each line will pass through a letter and a number. Write the letter above the appropriate number in the code at the top of the page.

$2\frac{1}{2} - 1\frac{7}{10}$	D	1	L
$2\frac{1}{4} - 1\frac{1}{2}$		4	
$1\frac{1}{3} - \frac{1}{2}$		3	A
$1\frac{2}{5} - \frac{1}{2}$			I
$2\frac{1}{6} - 1\frac{1}{2}$		6	F
$3\frac{1}{2} - 1\frac{3}{4}$		R	5
$1\frac{3}{10} - \frac{4}{5}$		U	2
$2\frac{1}{6} - \frac{2}{3}$		N	7
$1\frac{5}{8} - \frac{3}{4}$		9	8
			E

$\frac{3}{4}$	
$\frac{7}{8}$	
$\frac{9}{10}$	
$\frac{2}{3}$	
$\frac{4}{5}$	
$\frac{1}{2}$	
$1\frac{3}{4}$	
$1\frac{1}{2}$	
$\frac{5}{6}$	



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WHAT HAPPENED WHEN THE DENTIST AND THE MANICURIST HAD AN ARGUMENT?

To solve this riddle, factorize each of the expressions below. Rule a line to connect the trinomial in the left hand column with its factors in the right hand column. Each line will pass through a letter and a number. Write the letter above the appropriate number in the code.

$x^2 + 3x + 2$	■	F	■ $(x-6)(x+4)$
$x^2 - 8x + 15$	■	13	■ $(x-2)(x-2)$
$x^2 + 3x - 18$	■	6 A	■ $(x+6)(x-3)$
$x^2 - 3x - 4$	■	I 2	■ $(x+1)(x+2)$
$x^2 + 9x + 14$	■	H D 11	■ $(x-5)(x+4)$
$x^2 + 4x - 5$	■	8 U	■ $(x+3)(x+5)$
$x^2 - x - 20$	■	O 4 T	■ $(x+5)(x-1)$
$x^2 + 8x + 15$	■	3	■ $(x-4)(x+1)$
$x^2 - 7x + 12$	■	12 7	■ $(x+4)(x+2)$
$x^2 - 2x - 24$	■	1 Y 9 G	■ $(x-3)(x-1)$
$x^2 - 4x + 3$	■	E 5 10	■ $(x-4)(x-3)$
$x^2 - 4x + 4$	■	N L	■ $(x-3)(x-5)$
			■ $(x+7)(x+2)$

4	2	5	9	13	8	3	11	2	4

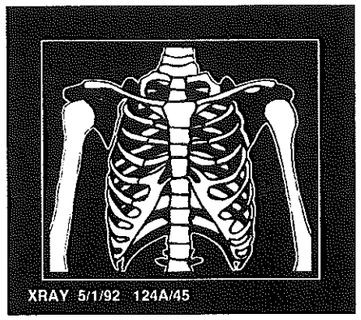
4	8	8	4	2	6	1	12	1	6	7	10

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HOW DID THE SKELETON KNOW IT WAS RAINING?

Simplify the following binomial products. Look for each simplified form and notice the letter next to it. Write that letter above the number of each question in the code below.

1. $(x + 2)(x + 3)$
2. $(x - 3)(x - 5)$
3. $(x + 2)(x - 6)$
4. $(x + 3)(x + 4)$
5. $(x + 5)(x - 2)$
6. $(x - 4)(x - 5)$
7. $(x + 1)(x + 4)$
8. $(x + 6)(x - 4)$
9. $(x - 2)(x - 1)$
10. $(x + 3)(x - 5)$
11. $(x - 2)(x - 2)$
12. $(x + 3)(x - 1)$
13. $(x + 5)(x + 2)$



U	$x^2 + 5x + 4$
D	$x^2 + 5x + 6$
T	$x^2 + 7x + 10$
H	$x^2 + 7x + 12$
F	$x^2 - 4x + 4$
B	$x^2 - 8x + 15$
I	$x^2 - 9x + 20$
C	$x^2 - 3x + 2$
O	$x^2 + 2x - 3$
L	$x^2 + 3x - 10$
E	$x^2 + 2x - 24$
S	$x^2 - 2x - 15$
N	$x^2 - 4x - 12$



4	8	9	12	7	5	1	11	8	8	5

6	13	6	3	4	6	10	2	12	3	8	10

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WHICH TUBA DO MUSICIANS HATE TO PLAY?

INSTRUCTIONS:
Simplify the following products of fractions. Look for your answers in the right hand column. Rule a line to join the question with its answer. Each line will pass through a number and a letter. Write the letter above the appropriate number in the code at the bottom of the page.



$\frac{1}{2}$	x	$\frac{3}{4}$	◆	◆ $\frac{1}{2}$
$\frac{2}{3}$	x	$\frac{3}{5}$	◆	◆ $\frac{1}{5}$
$\frac{3}{4}$	of	$\frac{2}{3}$	◆	◆ $\frac{3}{10}$
$\frac{1}{2}$	of	$\frac{2}{3}$	◆	◆ $\frac{2}{5}$
$\frac{2}{5}$	x	$\frac{1}{3}$	◆	◆ $\frac{2}{15}$
$\frac{1}{2}$	of	$\frac{2}{5}$	◆	◆ $\frac{1}{6}$
$\frac{3}{5}$	of	$\frac{5}{7}$	◆	◆ $\frac{3}{8}$
$\frac{2}{3}$	x	$\frac{1}{4}$	◆	◆ $\frac{3}{7}$
$\frac{4}{5}$	of	$\frac{3}{8}$	◆	◆ $\frac{1}{3}$



2	1	4	8	2	1	9	9	1	6	5	2	7	1	3

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WHAT DO YOU GET IF YOU CROSS A SHEEP WITH A STORM?



6	2	8	3			
4	1	6	7	5	8	3

Simplify the following products of mixed numerals. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle above.

CODE KEY

- $1\frac{1}{8}$ – B
- $1\frac{3}{4}$ – L
- $2\frac{2}{5}$ – E
- 4 – A
- $2\frac{1}{4}$ – T
- $1\frac{3}{5}$ – W
- $\frac{5}{6}$ – N
- $4\frac{1}{2}$ – K

1. $1\frac{1}{2} \times 1\frac{1}{6}$
2. $1\frac{1}{3} \times 1\frac{1}{5}$
3. $(1\frac{1}{2})^2$
4. $\frac{3}{4} \times 1\frac{1}{2}$
5. $\frac{3}{4} \times 6$
6. $1\frac{3}{5} \times 2\frac{1}{2}$
7. $\frac{2}{3} \times 1\frac{1}{4}$
8. $1\frac{1}{2} \times 1\frac{3}{5}$



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WHEN DID THE MAGICIAN GO TO SEE HIS FRIEND IN HOSPITAL?



INSTRUCTIONS:
Simplify each of the following binomial products. Rule a line to join each product in the left hand column with its simplified form in the right hand column. Each line will pass through a number and a letter. Write the letter above the number at the bottom of the page.

$(x + 3)(x + 6)$	$x^2 + 8x + 7$
$(x + 2)(x + 2)$	$x^2 - 2x - 24$
$(x + 7)(x + 1)$	$x^2 + 9x + 18$
$(x + 1)(x + 5)$	$x^2 - 11x + 30$
$(x - 3)(x - 4)$	$x^2 - 7x + 12$
$(x - 2)(x - 5)$	$x^2 + 4x + 4$
$(x - 1)(x - 6)$	$x^2 + 4x - 12$
$(x - 5)(x - 6)$	$x^2 - 7x + 6$
$(x + 6)(x - 2)$	$x^2 + 2x - 15$
$(x + 5)(x - 3)$	$x^2 + 6x + 5$
$(x - 6)(x + 4)$	$x^2 - 6x - 16$
$(x - 8)(x + 2)$	$x^2 - 7x + 10$

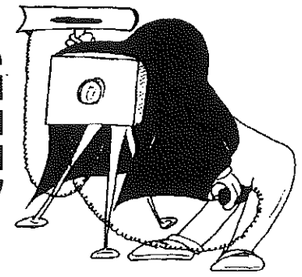
H 8 G 3 A 4 5 7 10 D R S W 11 U 12 I 9 1 Z 2 N 6 O

10	12	1	11	9	8	7	11	2	5	1	10	11	9	8

3	6	12	1	4

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WHAT DO YOU GET IF YOU CROSS A CAMERA WITH A MIRROR?



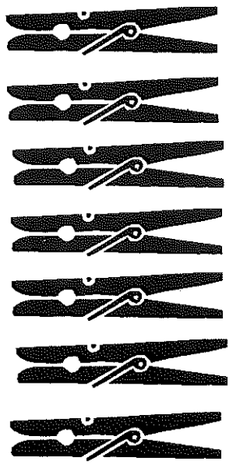
Instructions:
Simplify the expressions in the left hand column below. Rule a line to connect each expression with its simplified form in the right hand column. Each line will pass through a number and a letter. Write that letter above the appropriate number in the code. Keep working until you have the answer to the riddle.

2(x + 4)	♣	10m – 6
3(y + 5)	♣	5x + 5
2(3x + 7)	♣	2x + 8
5(2x + 3y)	♣	10x + 3
x(w – 2y)	♣	5x – 4
y(2x – m)	♣	3y + 15
4(3w – 1)	♣	wx – 2xy
2(5m – 3)	♣	8y – 2
3x(2 + w – m)	♣	6x + 14
2(x + 1) + 3(x – 2)	♣	7y – 3
4(y – 1) + 2(2y + 1)	♣	5x + 10
3(2 + x) + (4 + 2x)	♣	12w – 4
(3 + 4x) + 2(3x)	♣	10x + 15y
2(2y + 3) + 3(y – 3)	♣	6x + 3wx – 3mx
(5 – x) + 4x + 2x	♣	2xy – my

6	15	6	12	11	9	6	10	2	6	10			
			10	6	8	11	4	14	13	15	10	3	9	11	4	
						1	7	13	10	4	11	5	7			

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WHAT NAME IS GIVEN TO A PIG'S LAUNDRY?



INSTRUCTIONS:
Simplify each of the following. Look for your answers in the grid below. Cross out each box containing a correct answer. When you finish there will be 7 boxes remaining. Rearrange the letters in those boxes according to numerical value to give you the answer to the riddle.

$1\frac{1}{2} \div \frac{3}{4}$

$6 \div 1\frac{1}{2}$

$1\frac{1}{3} \div 2$

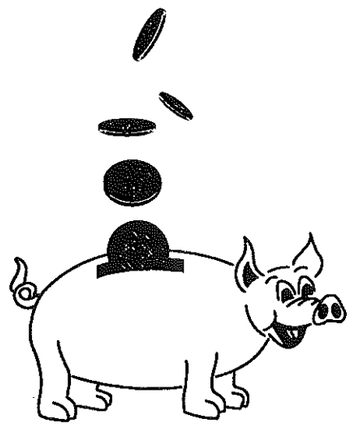
$3 \div \frac{2}{3}$

$\frac{5}{6} \div \frac{1}{2}$

$\frac{3}{4} \div 3$

$\frac{9}{10} \div \frac{3}{5}$

$\frac{3}{5} \div \frac{3}{4}$



4 – R $4\frac{1}{2}$	7 – H $1\frac{3}{4}$	2 – A $\frac{1}{4}$	3 – N 2	5 – A $2\frac{1}{2}$
2 – O $2\frac{2}{3}$	7 – M $\frac{2}{3}$	6 – S 3	5 – E $\frac{4}{5}$	1 – H $1\frac{1}{4}$
1 – S $1\frac{2}{3}$	4 – W $1\frac{3}{5}$	6 – T 4	3 – G $1\frac{4}{5}$	5 – O $1\frac{1}{2}$

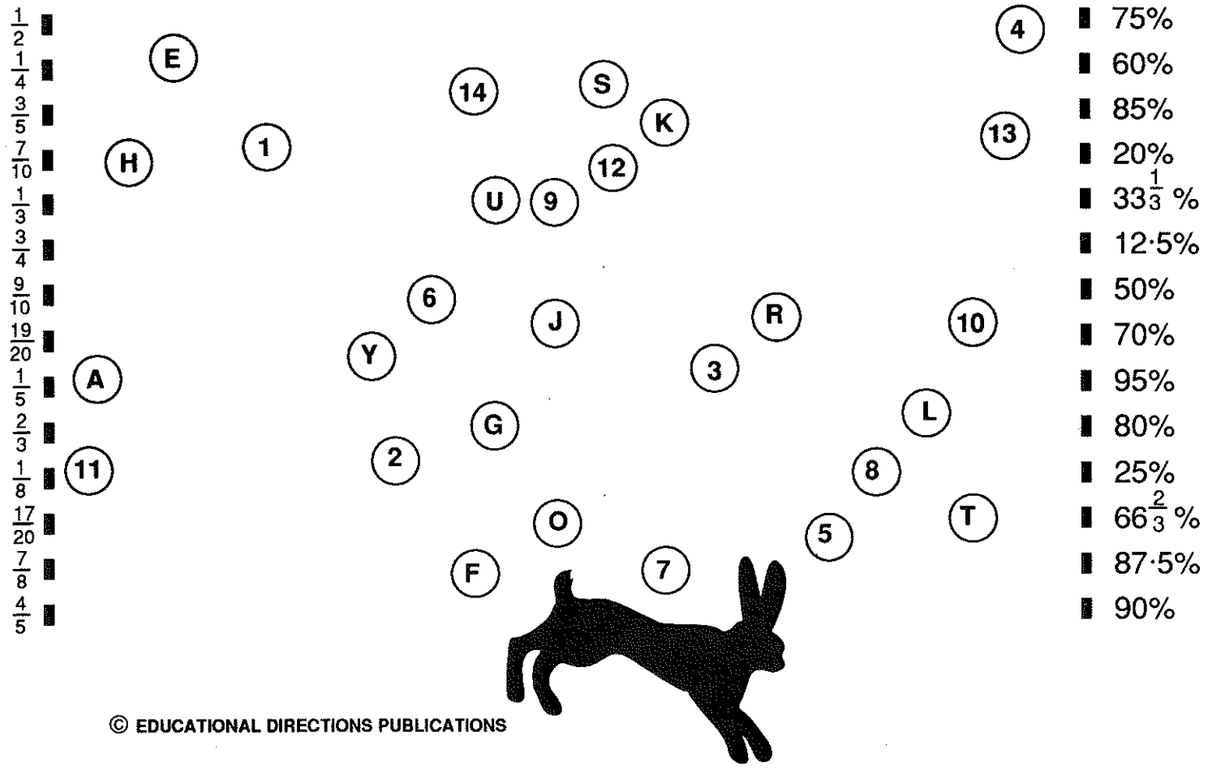
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HOW CAN YOU TELL WHICH RABBITS ARE THE OLDEST IN A GROUP?

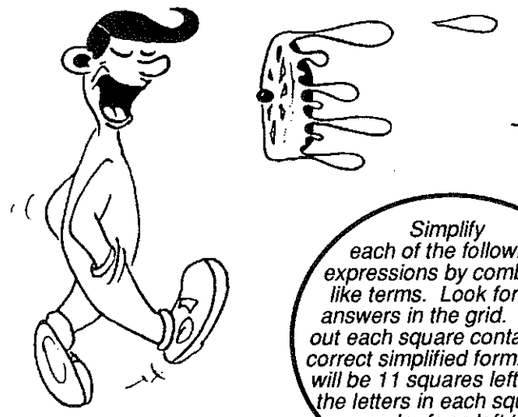
INSTRUCTIONS:
Match each of the fractions below with their equivalent percentages in the right hand column. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the number of the question in the code above.

	13
	9
	14
	2
	1
	8
	8
	4
	7
	8
	11

	2
	10
	12
	5
	11
	12
	3
	10
	6
	11
	12
	14



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WHAT DO YOU CALL A PERSON WITH A JELLY IN ONE EAR AND CUSTARD IN THE OTHER?

Simplify each of the following expressions by combining like terms. Look for your answers in the grid. Cross out each square containing a correct simplified form. There will be 11 squares left. Write the letters in each square in order from left to right.

- 2x + x + 3
- 3x + 4 + 5x + 2
- 2x – 4x + 3 + 4
- 2x + 2y – 5x – 6y
- 6 + 3y + 4y + 2y
- 6y – 4y + 2 + 7
- 2 – 7 + 3y – y
- y + 3y – 2y – 4y
- 6y + 5x – 2x + y
- 3x + 2y – y + x
- 4x + 3x – 5 + x
- 7 – 3x + 7x – 6
- 5 + 4y – 6y – 2
- 3y – 7y + y + 4

S 2y + 9	A 10 – 3x	M 4x + y	T 2x + 6	R 4x + 5
E 8x + 6	L 4x + 1	I 5 – 2x	N – 2y	O 3x + 3
F 6 + 7y	R 4 – 3y	E 2y – 5	A –3x – 4y	L 4x + 3y
M 3 – 2y	E 2x + 5	D 5 + 4y	G 7 – 2x	E 2x + 2y
A 4y	H 9y + 6	T 8x – 5	F 9x – 3y	S 3x + 7y

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WHY DID THE ORCHESTRA HAVE BAD MANNERS?

11.5 cm	1.5 km	9 cm	11.5 cm	9 cm	4 cm	1.5 km
5 m	4 cm	2 cm	10 m	500 m	2 cm	10 m
1.5 km	2 cm	18 m	2 cm	4 cm	9 cm	1.5 cm	18 m	1.5 km
11.5 cm	1.5 km	3.5 km	18 cm	3 m	1 875 m	

Look for your answers to the following questions in the code above.
Write the letter of each question above its answer.

A map is drawn with a scale of 1:100

What real-life distance is represented by a map measurement of:

L 3 cm?
K 5 cm?
W 10 cm?
C 18 cm?

What distance on the map would be needed to represent real-life distances of:

O 200 cm?
D 9 m?
I 11.5 m?



A map is drawn using a scale of 1:25 000

What real-life distance is represented by a map measurement of:

H 2 cm?
T 6 cm?
F 7.5 cm?
S 14 cm?

What distance on the map represents real-life distances of:

U 375 m?
N 1 km?
E 4.5 km?

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HOW DID THE BLIND CARPENTER REGAIN HIS SIGHT?

95%	25%	100%	42%	65%	48%	25%	66%

30%	100%	60%	95%	60%	90%	90%	25%	57·5%

60%	75%	66%	89%	60%	50%

Write each of the following test scores as percentages.

LOOK for your answers in the code above.

Write the letter of each question above its corresponding percentage.

E 12 out of 48

K 12 out of 25

R $11\frac{1}{2}$ out of 20

C 26 out of 40

U 9 out of 30

I 84 out of 200

W 10 out of 20

D 33 out of 50



A 6 out of 10

N 39 out of 52

S 89 out of 100

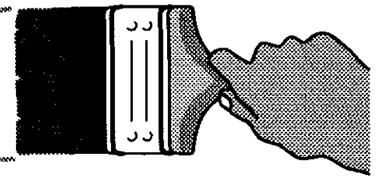
M 36 out of 40

H 19 out of 20

P 40 out of 40

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WHAT IS ANOTHER NAME FOR A GRANDFATHER CLOCK?



Convert each of the fractions below to percentages, correct to one decimal place. Look for your answers in the code below. Write the letter of each question above its answer.

M $\frac{9}{16}$

R $\frac{5}{6}$

O $\frac{23}{40}$

A $\frac{5}{8}$

D $\frac{11}{12}$

I $\frac{4}{7}$

T $\frac{1}{3}$

N $\frac{7}{16}$

E $\frac{9}{22}$

L $\frac{2}{3}$

62·5%	43·8%	57·5%	66·7%	91·7%

33·3%	57·1%	56·2%	40·9%	83·3%

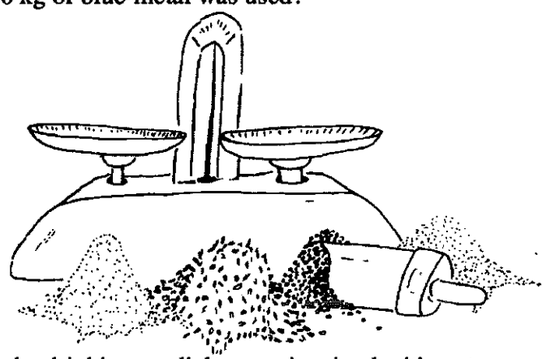
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WHEN A BAKER WANTS TO SLEEP, WHAT SIGN DOES SHE PUT UP?

6	7	10	9	11	4	10	8	6	1	3	8	10	5	2

Solve each of the following. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of the question in the puzzle above.

1. In a science class the ratio of girls to boys is 4:3. If there are 12 girls, how many boys are there?
2. If it takes 720 grams of flour to make 4 cakes, how much flour would be required to make 5 cakes?
3. Powdered milk is mixed with water in the ratio of 1:3 to make milk suitable for drinking. How many cups of water should be added to 2 cups of powdered milk?
4. A chef needs 3 eggs to make 2 omelettes. How many omelettes could be made from a dozen eggs?
5. Concrete is mixed using 3 parts of sand, 2 parts of blue-metal and one part of cement. How much sand would be required if 350 kg of blue-metal was used?

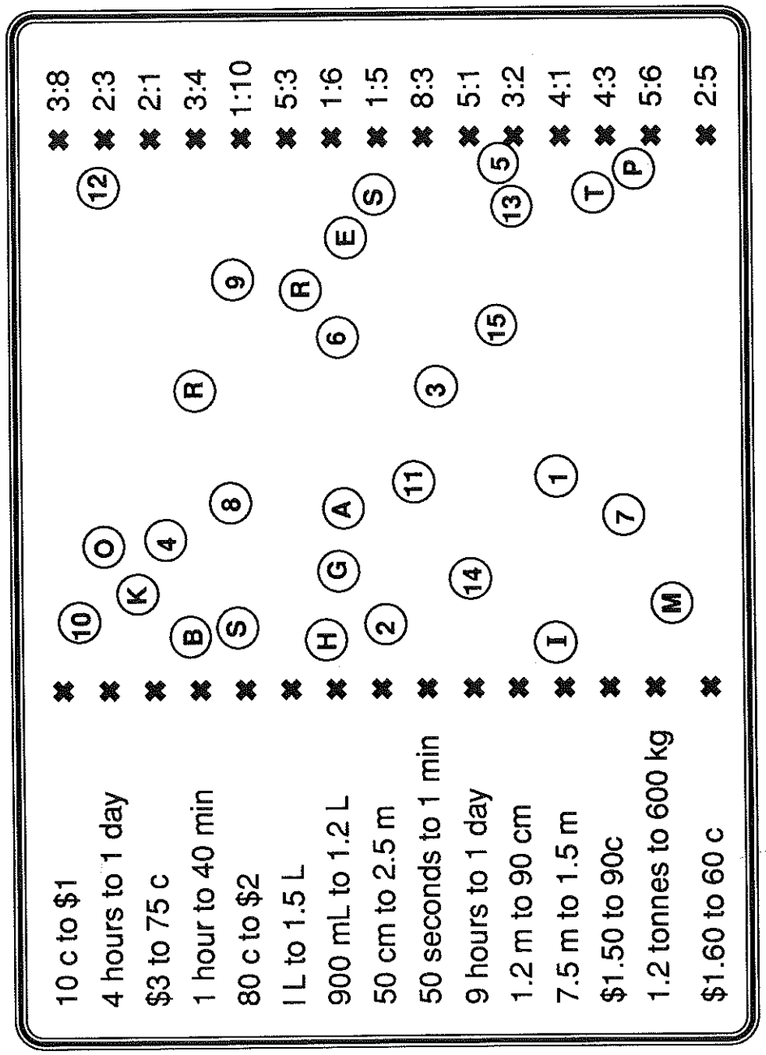


CODE KEY

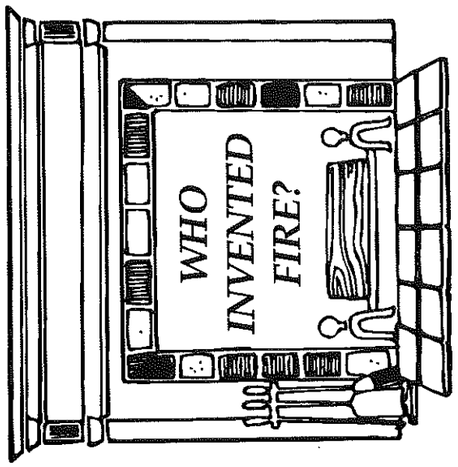
8	– N
25 kg	– H
27	– O
9	– I
7·2 L	– D
7·5 L	– U
525 KG	– R
24 L	– G
6	– S
36	– T
900 g	– B

6. To make drinking cordial, syrup is mixed with water in the ratio 2:12. How much water would need to be added to 1.2 litres of syrup?
7. The ratio of dogs to cats at a pet haven was 3:5. If there were 45 cats, how many dogs were there?
8. Four loaves of bread are required to make lunch for 24 hungry children. At the same rate, how many could be fed with 6 loaves?
9. Grow-Fast liquid fertilizer is made by mixing liquid concentrate with water in the ratio 1:400. How much water should be mixed with 60 mL?
10. Green paint is made by mixing blue paint with yellow paint in the ratio 4:5. How much yellow paint should be mixed with 6 litres of blue paint?
11. A food caterer requires 2 kg of meat for every 12 people. How much meat would be required to cater for a party of 150 people?

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	4
	6
	12
	14
	5
	1
	9
	11
	15
	8
	13
	3
	7
	10
	2



Match each ratio in the left hand column with its simplified form in the right hand column. Rule a line to join each pair. Each line will pass through a number and a letter. Write the letter above the appropriate number in the code at the bottom of the page.

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WHAT DO YOU CALL A ROW OF RABBITS WALKING BACKWARDS?

Answer each of the following questions. Look for your answers in the Code Key. Notice the letter next to each answer. Write that letter above the number of each question in the puzzle below.

1. In a sale, a coat is advertised with a 20% discount. What is that discount as a fraction?
2. Jason scored 27 out of 40 on his recent science test. What is this mark as a percentage?
3. If a client paid her bill within a month, she would have the bill reduced by $\frac{1}{10}$. What percentage reduction would she get?
4. To support a local charity, MacScotties will donate 60 c for each $5 hamburger that is sold. What percentage of the cost of each burger will be donated?
5. Each MacScotties hamburger is $\frac{3}{5}$ beef by weight. What percentage of the weight of each burger is beef?
6. In a class of 32 students, 20 were girls. What fraction of the class was girls?
7. In a packet of 60 jellybeans, 9 were black? What percentage of the jellybeans was black?
8. In a multiple choice test, Tanya answered 75% of the questions correctly. What fraction of the questions did she get right?
9. A carton of fruit juice cordial advertises that it contains 35% orange juice. What fraction of the drink is orange juice?
10. After a margarine taste-test, it was claimed that 19 out of every 20 people preferred brand X. What percentage of the people tested preferred brand X?

CODE KEY

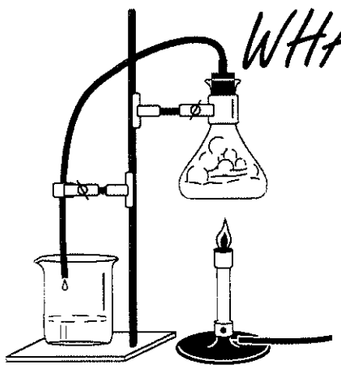
10%	– D
95%	– A
15%	– C
$\frac{1}{5}$	– G
12%	– H
$\frac{3}{4}$	– L
60%	– I
$\frac{7}{20}$	– E
67·5%	– N
$\frac{5}{8}$	– R

10	6	9	7	9	3	5	2	1

4	10	6	9	8	5	2	9

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WHAT LARGE ANIMAL ALWAYS TAKES MEDICINE?



63	4	7	6	2	32	5	54	16	18	30	20	25	80	3

Find the value of the variable in each pair of equivalent ratios. Look for your answer in the code. Write each letter above its answer. Keep working until you have solved the riddle.

2:5 = P:15

3:4 = 12:O

5:9 = 30:H

H:1 = 16:4

5:6 = I:30

1:C = 5:15

3:8 = 30:A

5:7 = 45:A

4:3 = R:15

3:C = 18:30

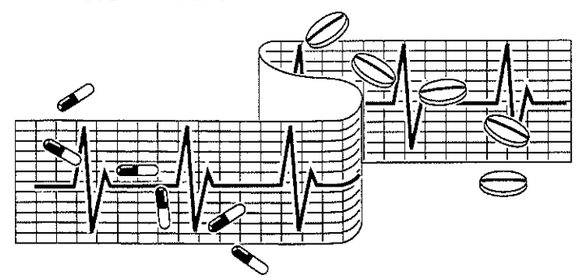
1:6 = 5:D

P:9 = 4:18

I:8 = 70:80

9:10 = N:20

5:4 = 40:O



WHAT DO YOU GET IF YOU CROSS A COW WITH A CAMEL?

Each of the ratios in the left hand column below can be simplified. Look for the simplified form in the right hand column. Rule a line to match each pair. Each line will pass through a number and a letter. Write the letter above the appropriate number in the code.

8	2	6	4	5	3	7	1	9	15	11	12	13	14	10

8:10 ✇
12:21 ✇
12:4 ✇
21:14 ✇
6:12 ✇
12:20 ✇
30:6 ✇
12:10 ✇
8:40 ✇
12:18 ✇
40:30 ✇
45:6 ✇
10:16 ✇
6:15 ✇
14:20 ✇

2:3 ✇
4:7 ✇
7:10 ✇
1:2 ✇
15:2 ✇
4:5 ✇
4:3 ✇
2:5 ✇
3:1 ✇
6:5 ✇
3:2 ✇
1:5 ✇
5:1 ✇
3:5 ✇
5:8 ✇

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